

TASK-EFFICIENT AGENT EVALUATION USING CACHED TRACES

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ABSTRACT

Efficient benchmarking estimates a system’s full-benchmark performance from a small number of task executions. Standard approaches rely solely on the binary correctness of executions from a collection of already-evaluated reference systems. An agentic system, however, produces a rich trace for each execution that can reveal substantially more about its capabilities. Exploiting this information requires useful trace representations. We propose representing a trace by extracting a structured, rubric-guided summary of it with an LLM and embedding the summary with a standard text encoder. These representations preserve correctness- and task-related signal while minimizing the presence of system identity-, model family-, and harness-related signals that dominate raw trace embeddings. Given this representation for each task execution from all reference systems, we construct low-dimensional representations of the systems, predict a new system’s score from its position among the reference systems, and combine the resulting geometric prediction with a two-parameter logistic item-response model. On SWE-bench Verified and Terminal-Bench 2.0, the proposed estimator reduces mean absolute error from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$), respectively, when provided a single random executed task relative to an item-response model alone. These results indicate that trace geometry and correctness outcomes carry complementary predictive signal and enable more accurate benchmark score estimation from limited agent executions.

1 INTRODUCTION

The dominant paradigm of generative model use is increasingly agentic: systems plan, invoke tools, observe intermediate results, and iterate over long horizons to complete complex tasks (Yang et al., 2024; Wang et al., 2025). As agentic capabilities increase, so does the range of settings in which a given system might be deployed – and, with it, the number of benchmarks on which the system must be evaluated before it can be trusted with consequential work (Kapoor et al., 2026). Agentic evaluation, however, is expensive as each task execution requires a properly sandboxed environment and a single task execution routinely consumes hundreds of thousands of tokens (Jimenez et al., 2024; Terminal-Bench Team, 2026) and can regularly exceed \$2,000 (Kapoor et al., 2026). Given this cost, efficient evaluation is necessary to increase the evaluation coverage of an agentic system.

Task-efficient evaluation methods (e.g., benchmark compression and anchor-item selection (Vivek et al., 2024; Maia Polo et al., 2024), item response theory and adaptive testing (Kipnis et al., 2025; Truong et al., 2025), leveraging cached responses (Helm et al., 2026a), etc.) address this problem by finding structure in item-level information available from previously-evaluated reference systems to decrease the cost of evaluating a new system. Some task-efficient evaluation methods require only the item-level scores from reference systems (Vivek et al., 2024; Kipnis et al., 2025). Others leverage the content of cached artifacts, such as natural language responses or residual streams, to measure the similarity of systems on a set of tasks (Duderstadt et al., 2023; Helm et al., 2026b;a). Notably, agentic workflows often produce artifacts when executing a task – file diffs, tool calls, intermediate plans, and error messages – recorded together in an execution trace that can be used to this effect. Unfortunately, existing methods for using cached artifacts rely on off-the-shelf embedding functions that are not immediately well-suited for representing traces.

We make two primary contributions:

Table 1: Mean Absolute Error (MAE) of benchmark score estimators for SWE-bench Verified and Terminal-Bench 2.0 under leave-one-family-out evaluation and different task selection protocols (random & adaptive). **Bolded** values are the best method for a given benchmark, task selection, and number of tasks. Underlined values are statistically significant ($\alpha = 0.05$) comparisons to IRT per the two-sided Wilcoxon signed-rank test. The TRUBRIC and IRT blend – which takes advantage of both trace geometry and per-execution score – is the best or near-best in all settings.

Num. tasks m	SWE-bench Verified						Terminal-Bench 2.0					
	random			adaptive			random			adaptive		
	1	5	20	1	5	20	1	5	20	1	5	20
sample score	0.442	0.175	0.082	0.380	0.151	0.072	0.396	0.164	0.073	0.360	0.124	0.132
IRT	0.127	0.088	0.048	0.080	0.062	0.034	0.137	0.097	0.052	0.112	0.067	0.039
raw trace	<u>0.099</u>	0.093	0.089	0.096	0.083	0.081	0.135	0.126	0.120	0.128	0.125	0.113
generic	<u>0.079</u>	<u>0.071</u>	0.068	0.074	0.070	0.069	0.139	0.127	0.115	0.144	0.130	0.117
trubric	<u>0.081</u>	<u>0.069</u>	0.060	0.068	0.056	0.060	<u>0.127</u>	0.107	0.093	0.110	0.092	0.084
IRT + trubric	<u>0.081</u>	<u>0.063</u>	<u>0.043</u>	<u>0.062</u>	<u>0.048</u>	<u>0.034</u>	<u>0.122</u>	<u>0.090</u>	<u>0.050</u>	<u>0.099</u>	<u>0.059</u>	<u>0.037</u>

1. **TRUBRIC trace representations.** We introduce TRUBRIC (*trace rubric*): a trace representation constructed by extracting a task specific structured summary of each trace with an LLM and embedding the summary with a standard text encoder. We show that TRUBRIC embeddings preserve information relevant to the task and behavior of the system while minimizing system-, model family-, and harness- specific signal across seven modern text encoders.

2. **Task-efficient evaluation using cached traces.** Given the TRUBRIC embeddings from a set of tasks for a target system and a collection of reference systems, we evaluate their utility for predicting the target system’s full-benchmark score on SWE-bench Verified and Terminal-Bench 2.0 under three different agentic system evaluation protocols. We find that TRUBRIC-based estimators, combined with an item-response model, reduce single-task mean absolute error from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$) on the two benchmarks, respectively, and are nearly always better than baselines at any task budget (Table 1).

1.1 PROBLEM STATEMENT

For our purposes, an agentic system $f \in \mathcal{F}$ is a random mapping from a task space $\mathcal{Q}$ to a trace space $\mathcal{X}$. A benchmark is a fixed collection of tasks $Q^* = \{q_1, \dots, q_M\} \subset \mathcal{Q}$. Each trace is evaluated by the benchmark’s trace-level scoring function $s : \mathcal{X} \rightarrow \{0, 1\}$ and the system is evaluated by its benchmark score $y(f) = \frac{1}{M} \sum_{j=1}^M \mathbb{E}[s(f(q_j))]$ where the expectation is with respect to the randomness of f . We write y for $y(f)$ when the context is clear.

When evaluating a new system, we typically have access to $\mathcal{D} = \{\{(x_{ij}, y_{ij})\}_{j=1}^M\}_{i=1}^n$, the traces and trace-level scores of n previously-evaluated reference systems $f_1, \dots, f_n$, where $x_{ij} \sim f_i(q_j)$, $y_{ij} = s(x_{ij})$ and $y_i = y(f_i)$. For the target system f_0 , we observe traces (and, optionally, trace-level scores) on only a subset $Q \subset Q^*$ of $|Q| = m \ll M$ tasks and our goal is to estimate the system’s full benchmark score y_0 using only the target’s m executions and $\mathcal{D}$.

2 METHODS

We first present TRUBRIC and then describe its role in geometry-based efficient benchmarking.

2.1 TASK-SPECIFIC RUBRICS FOR REPRESENTING TRACES

A trace representation for benchmark prediction should place two traces close together when they reflect similar execution behavior on the task – and should do so regardless of which harness, model family, or specific system produced them. TRUBRIC constructs such a representation in two stages.

First, for each task q_j , an LLM is prompted with the task description to instantiate a *task-specific rubric*: a set of K fields $R_j = (r_{j1}, \dots, r_{jK})$, where each field describes one aspect of execution behavior to attend to when reading a trace from that task (e.g., how the relevant code was localized,

how the fix was validated, how errors were handled). Field names are shared across tasks while their descriptions are instantiated per task; rubrics are generated once per task and shared across all systems, so their cost is amortized over the reference collection. We use $K = 6$ fields by default and study the effect of rubric size in Section 3.

Second, for each trace x_{ij} , an LLM produces a structured summary $z_{ij} = (z_{ij}^1, \dots, z_{ij}^K)$, where z_{ij}^k is a natural-language description of the trace with respect to field r_{jk} . Each field is embedded separately with an off-the-shelf text encoder g and the embeddings are concatenated, yielding the trace representation $\phi(x_{ij}) = (g(z_{ij}^1), \dots, g(z_{ij}^K))$. Finally, representations are *consensus-centered*: for each task and field, the mean representation across systems is subtracted, removing content common to every attempt of the task (e.g., restatements of the problem) and emphasizing differences in behavior. We compare against two alternatives constructed with the same encoder: embedding the raw trace directly (head and tail slices of the rendered log) and summarizing with a *generic* rubric whose field descriptions are fixed across tasks.

To quantify what a representation encodes, we train class-balanced linear probes to predict five attributes of a held-out trace from its embedding: task and correctness (signal an evaluation should retain) and system identity, model family, and harness (metadata it should not). Probes for system-level attributes are evaluated on held-out tasks, and probes for task and correctness on held-out groups of systems, so that reported balanced accuracies reflect generalization rather than memorization.

2.2 INFERENCE IN THE PRODUCT KERNEL PERSPECTIVE SPACE

Given trace representations, we represent *systems* following the Product Kernel Perspective Space (PKPS) construction (Anonymous, 2026). Let ψ denote a text embedding of the task description and $\kappa_\sigma(q, q') = \exp(-\|\psi(q) - \psi(q')\|^2 / 2\sigma^2)$ a Gaussian kernel on tasks. For systems a and b observed on task sets Q_a and Q_b , define

$$\langle a, b \rangle = \frac{\sum_{q \in Q_a} \sum_{q' \in Q_b} \kappa_\sigma(q, q') \langle \phi(x_{aq}), \phi(x_{bq'}) \rangle}{\sum_{q \in Q_a} \sum_{q' \in Q_b} \kappa_\sigma(q, q')}, \quad d^2(a, b) = \langle a, a \rangle + \langle b, b \rangle - 2\langle a, b \rangle.$$

The kernel compares traces from *similar* – rather than only identical – tasks, so the target (observed on m tasks) can be compared to references (observed on all M) without discarding unpaired information; as $\sigma \rightarrow 0$ the construction recovers paired-only comparisons.

We consider two estimators of y_0 from this geometry. The first is distance-weighted k -nearest neighbors: $\hat{y}_{\text{geom}}$ is the weighted mean of the benchmark scores of the k references closest to the target. The second applies classical multidimensional scaling to the $(n+1) \times (n+1)$ distance matrix and fits a ridge regression from the resulting coordinates to the reference scores; reference predictions are made leave-one-out via the hat matrix so that they remain honest. All hyperparameters – the bandwidth σ , the choice of estimator, and its parameters – are selected by treating each reference system as a pseudo-target, predicting its score from the remaining references, and minimizing the mean reference error. No target labels are used at any point, so the geometric estimators apply even when the target’s executions cannot be graded.

As a score-only baseline and blending partner we use a two-parameter logistic (2PL) item-response model (Birnbau, 1968): $\mathbb{P}(y_{ij} = 1) = \text{logistic}(a_j(\theta_i - b_j))$ with item discriminations a_j and difficulties b_j fit on $\mathcal{D}$, the target ability θ_0 estimated from its m graded executions, and $\hat{y}_{\text{IRT}}$ given by the average predicted success probability over Q^* . The combined estimator is the convex blend $\hat{y} = \alpha \hat{y}_{\text{IRT}} + (1 - \alpha) \hat{y}_{\text{geom}}$, with $\alpha \in [0, 1]$ selected by the same reference-pseudo-target procedure. For adaptive task selection, tasks are chosen sequentially to maximize Fisher information about the target’s ability under the reference-fit 2PL model.

3 RESULTS

Rubric-based preprocessing produces more useful trace representations. We compare embeddings of raw traces with embeddings of structured summaries produced using either a generic rubric or a task-specific rubric, TRUBRIC. Figure 1 shows that both rubric constructions reduce recoverable information about system identity, model family, and harness while retaining task and

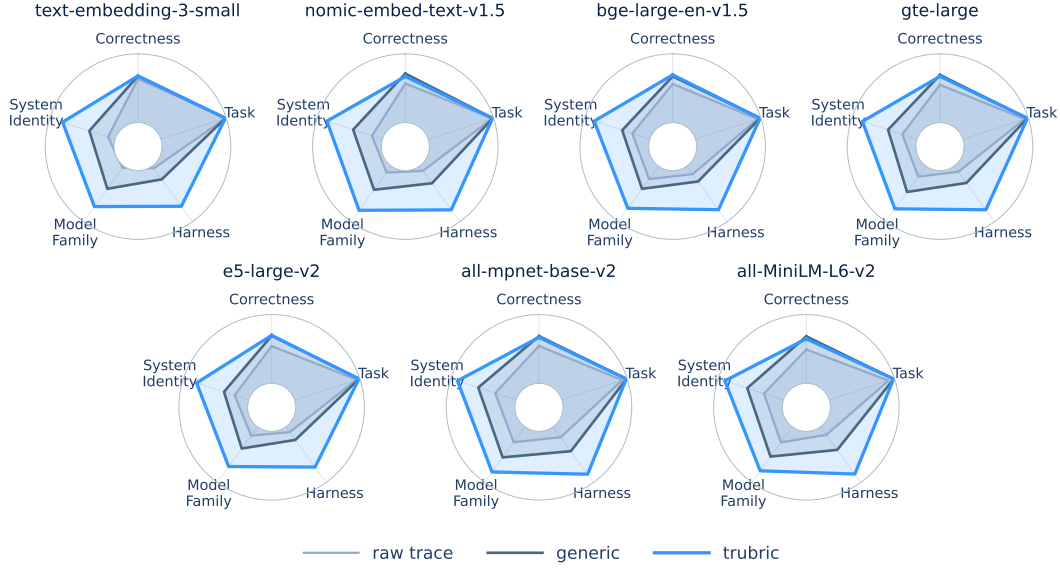


Figure 1: Comparisons of the properties of three different methods for embedding executed task traces across seven different off-the-shelf text encoders: embedding the raw trace, embedding a structured summary based on a generic rubric, and embedding a structured summary based on a task-specific rubric. As measured by the balanced accuracy of held-out linear probes, described in Section 2, the embedding of the task-specific summary is better at removing undesirable signal while preserving relevant signal across all seven encoders.

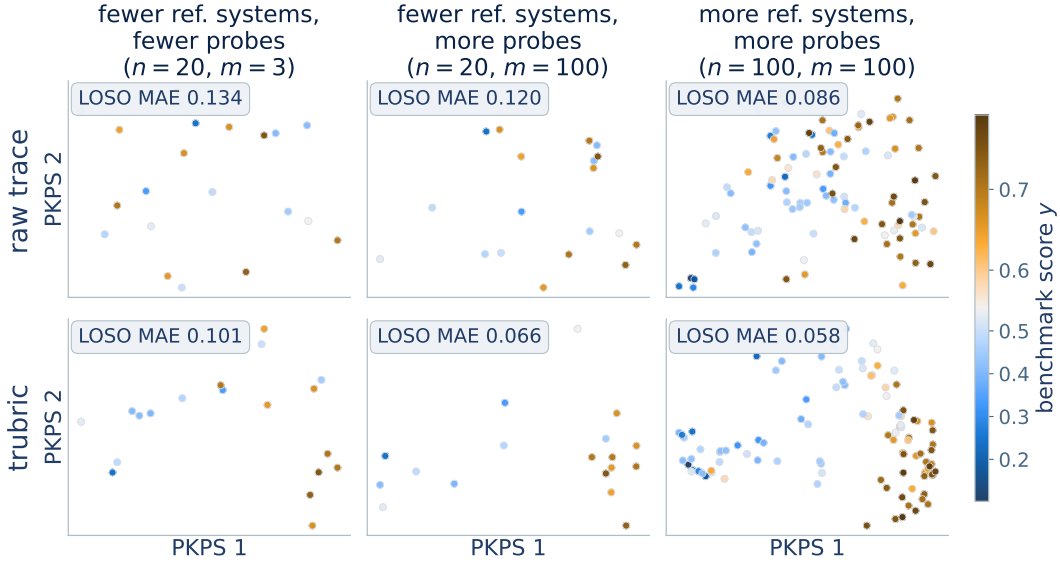


Figure 2: The 2-d Product Kernel Perspective Space (PKPS) induced with raw trace embeddings (top) and the TRUBRIC embeddings (bottom) under different number of reference systems and number of task executions. Each dot is a system colored by full benchmark score on SWE-bench Verified. The manifold of systems under the TRUBRIC embeddings is more strongly correlated with the full benchmark score in paired settings. We report the average Mean Absolute Error (MAE) under leave-one-system-out (LOSO) protocol to quantify this observation.

correctness information. As discussed in Section 1, these attributes can introduce nuisance variation into trace similarity, obscuring similarities in how systems execute tasks. Task-specific rubrics generally produce a larger reduction in identity-related signals than generic rubrics. This qualitative

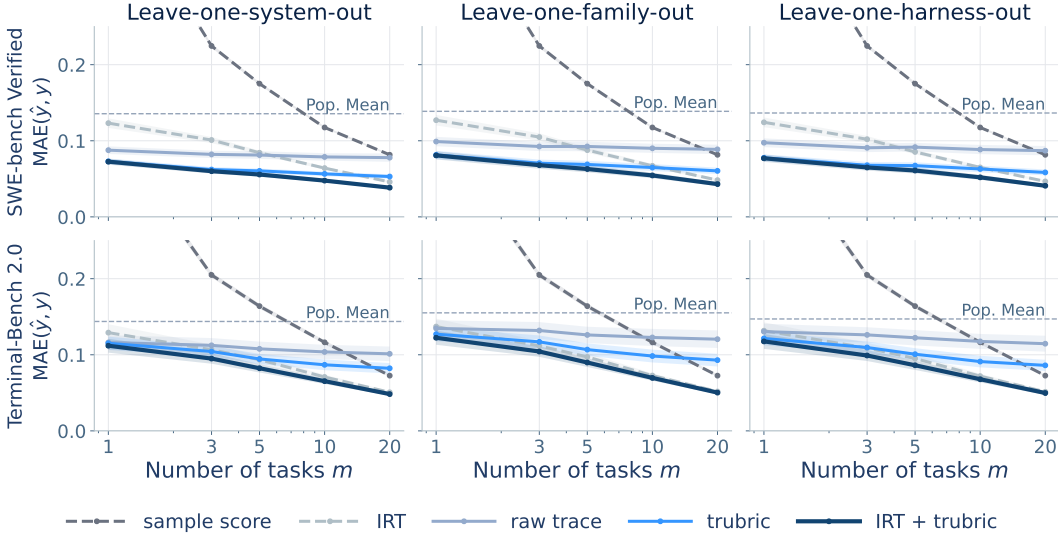


Figure 3: The Mean Absolute Error (MAE) of benchmark score estimators as a function of the number of executed tasks for SWE-bench Verified (top) and Terminal-Bench 2.0 (bottom) under three different realistic evaluation protocols: leave-one-system-out (left), leave-one-family-out (center), and leave-one-harness-out (right). Reported MAE is the average over 50 random subsets of tasks. Shaded areas represent ± 1 S.E. The TRUBRIC and IRT blend dominates all methods (i.e., is never worse and is sometimes better) across all protocols, datasets, and number of executed tasks.

pattern persists across seven embedding functions with little overall variation, although the precise magnitudes differ.

These differences are also reflected in the system-level geometry. Figure 2 shows a clearer relationship between system position in the embedding space and benchmark score for TRUBRIC than for raw-trace embeddings. In Figure 3, prediction error decreases more rapidly with additional tasks when using TRUBRIC representations than when using raw-trace embeddings. On SWE-bench Verified, increasing the task budget from 1 to 20 reduces geometry-only MAE from .081 to .060 for TRUBRIC, compared with .099 to .089 for raw traces. Table 1 further shows that generic rubrics are competitive on SWE-bench, particularly at small budgets, though task-specific rubrics provide larger improvements on Terminal-Bench 2.0.

Trace geometry alone supports task-efficient benchmark prediction. Table 1 and Figure 3 show that rubric-based trace geometry predicts full-benchmark performance more accurately than the sample score across small task budgets. Under leave-one-family-out evaluation, this advantage holds from 1 through 10 randomly selected tasks on both benchmarks. The geometric estimator also outperforms the population-mean baseline, indicating that its accuracy reflects useful differences between systems rather than just regularizing towards the average reference score. We emphasize that these geometric methods *only* require the target’s trace content, and *not* its correctness labels, which means that these methods are applicable in settings where grading (even a small number of) target tasks is infeasible (due to e.g. cost, lack of expertise, etc.), provided scored reference systems are available.

Trace geometry can also outperform IRT, which already exploits the correctness patterns of reference systems. On SWE-bench, geometry remains more accurate through 10 random tasks (Figure 3). On Terminal-Bench, geometry outperforms IRT at one task, but IRT is more accurate from three tasks onward. The advantage of trace geometry over outcome-only methods is largest when few target executions are available, while its relative performance at larger budgets varies by benchmark.

Combining trace geometry with correctness outcomes improves prediction beyond either source alone. The two estimators exhibit different behavior as the task budget grows: geometric predictions are informative with very few executions, while IRT improves substantially as the

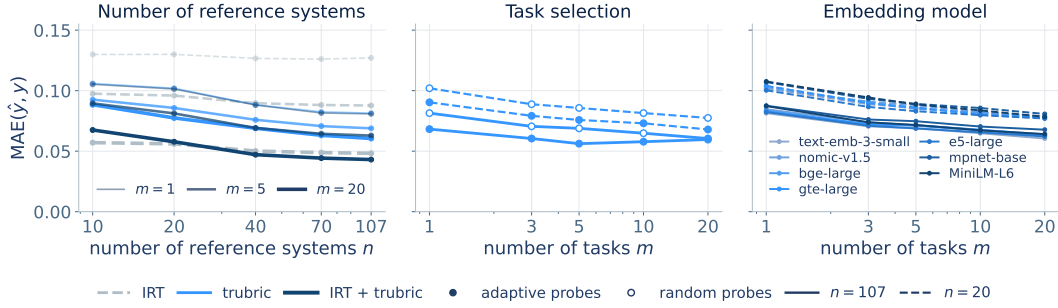


Figure 4: Mean Absolute Error (MAE) of benchmark score estimators with access to different number of reference systems (left), different task selection procedures (center), and using different text encoders (right) under the leave-one-family-out evaluation protocol. Reported MAE is the average over 50 random subsets of tasks; adaptive task paths are deterministic given the reference systems. Adaptive tasks are selected sequentially to maximize Fisher information under a two-parameter logistic model fit to the reference systems, as described in Section 2.

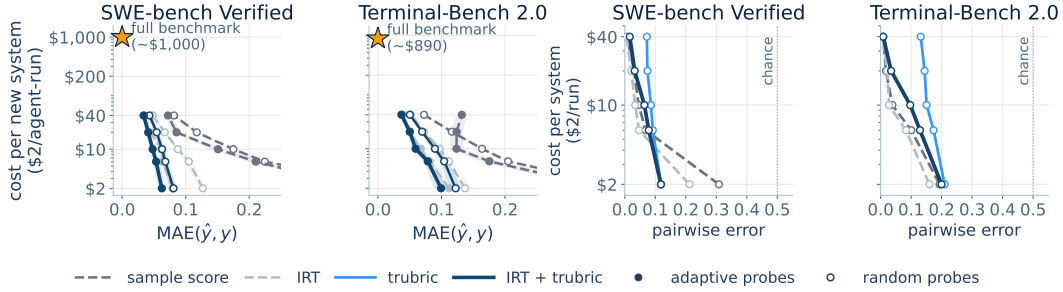


Figure 5: The cost of achieving a particular Mean Absolute Error (MAE; left and center left) and pairwise ranking error (center right and right) when using different benchmark score estimators on SWE-bench Verified and Terminal-Bench 2.0. Stars indicate the cost of producing the true benchmark score. An ideal method has low error (to the left) and is cheap (lower). Costs are calculated at a nominal \$2 per task execution. Values are estimated using the leave-one-family-out results underlying Table 1 and, for pairwise error, leave-two-out comparisons of system pairs whose true scores differ by at least 0.05.

number of tasks increases. We combine the geometric and IRT predictions using the weighting procedure described in Section 2, referring to the resulting estimator as IRT + TRUBRIC. Figure 3 shows that combining their predictions improves accuracy across evaluated budgets. At five random tasks on SWE-bench, IRT + TRUBRIC achieves .063 MAE, compared with .069 for geometry alone and .088 for IRT. On Terminal-Bench – where IRT is relatively stronger – the combination achieves .090, compared with .107 for geometry and .097 for IRT.

The improvement therefore extends beyond choosing whichever method performs better on a particular benchmark. These results support the complementary predictive value of execution content and correctness outcomes. The broad pattern persists across the system-, family-, and harness-exclusion protocols in Figure 3, although the size of the improvement varies. As Table 1 shows, IRT + TRUBRIC is best or near-best across the reported settings, though its advantage over IRT generally shrinks as the number of correctness observations increases.

Trace-informed estimation reduces the cost of reaching a given prediction accuracy. Figure 5 translates the error curves into execution costs, illustrating how more informative estimates can reduce the number of tasks required for a desired accuracy. For this comparison, we assume a cost of \$2 per agent execution. On SWE-bench, IRT + TRUBRIC reaches an MAE below .10 with one random task, compared with five tasks for IRT and twenty for the sample score – the smallest tested

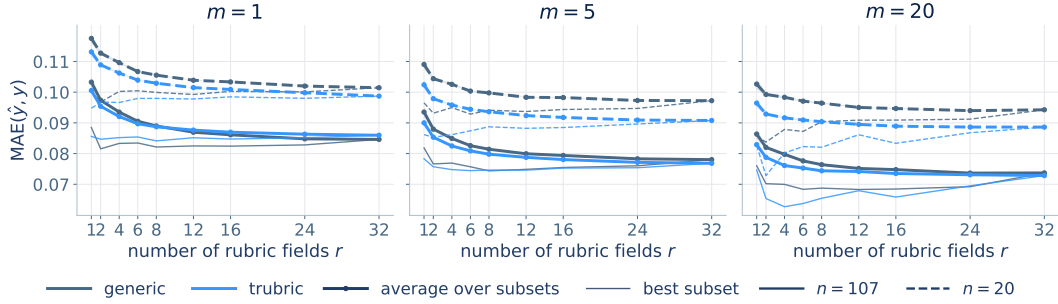


Figure 6: The effect of the number of fields present in the rubric used to extract a structured summary of the trace on Mean Absolute Error (MAE) for different number of executed tasks and different number of reference systems for both task-specific and generic rubrics. We report both the MAE for the average (bold) and best performing (thin) sets of fields. In general, a task-specific rubric is preferred to a generic rubric when there are fewer reference systems and more executed tasks available.

budgets meeting this threshold. This corresponds to estimated target execution costs of \$2, \$10, and \$40, respectively.

Combining trace information with adaptive sampling also reduces cost at a more stringent accuracy level: five adaptive tasks with IRT + TRUBRIC achieve approximately the same error as twenty random tasks with IRT, .048 MAE (Table 1), corresponding to \$10 rather than \$40. These comparisons concern the cost of executing tasks for a new system and assume an existing reference collection. We assume that reference executions are available from community archives or that their acquisition cost is amortized over a large number of target evaluations. In this calculation, we treat the cost of LLMs used to do (t)rubric summarization and the cost of API-based embedding functions as negligible compared to the cost of running the agents on the task. These comparisons illustrate how extracting additional information from each execution can make accurate benchmark estimation feasible at substantially smaller task budgets.

Performance improves with larger reference libraries and adaptive task selection. Figure 4a shows that increasing the number of reference models generally improves both IRT and geometry-based estimators. At five random tasks on SWE-bench, expanding the library from 20 to 107 systems reduces geometry-only MAE from .086 to .069 and IRT + TRUBRIC MAE from .081 to .063. The contribution of geometry depends on reference availability: given a small number of reference models and a large target-task budget, IRT can outperform geometry-based methods.

Adaptive task selection provides a further improvement. With five adaptive tasks, IRT + TRUBRIC achieves .048 MAE on SWE-bench, compared with .063 for five random tasks (Table 1). Together, Table 1 and Figure 4b show that adaptive selection improves both IRT and geometry-based prediction. These findings demonstrate that trace-based estimation can be used alongside adaptive sampling.

Figure 4c further shows that the qualitative behavior persists across seven embedding functions. On SWE-bench, the impact of encoder choice appears small compared with the effects of the number of target tasks and the number of reference models.

Broader rubrics improve average performance, though strong, small rubrics may exist. Figure 6 varies the number of fields included in generic and task-specific rubrics, averaging performance over subsets drawn from a bank of 32 fields. For both constructions, adding fields generally improves average prediction accuracy, with diminishing returns. At five tasks and the larger reference library, increasing a task-specific rubric from one to six fields reduces average MAE from .090 to .081; using all 32 fields reduces it further to .077. This pattern is consistent with an ensembling effect from combining descriptions of multiple aspects of execution. Task-specific rubrics generally outperform generic rubrics in this experiment, with small reversals at the one-task budget.

The thin curves show that some small subsets perform substantially better than the average subset of the same size. Each point selects the lowest average evaluation error among the tested subsets separately for that setting, averaging over probe draws and all 107 target systems even when the reference library contains 20 systems. These retrospective selections demonstrate the existence of favorable small rubrics on the evaluation panel, rather than a validated procedure for identifying them in advance. The selected subsets need not be nested as field count increases, so their performance need not improve monotonically.

4 RELATED WORK

Representing agentic traces. The standard tools for embedding model outputs are text encoders trained on sentence-level semantic similarity (Reimers & Gurevych, 2019; Su et al., 2023; Nussbaum & Duderstadt, 2025). Agentic traces, however, implicitly or explicitly encode system metadata that may not be relevant for downstream tasks (Sun et al., 2025; Oderinwale, 2026) and are not necessarily encoded well by semantic embedding models. Instead, traces have primarily been studied through taxonomies of failure modes and actions (Cemri et al., 2025; Shu et al., 2026) and LLM judges of agent behavior (Zhuge et al., 2025; Mehtiyev & Assunção, 2026). Parallel work uses LLMs to adapt model outputs to be more naturally compared via their semantics (Ravfogel et al., 2024; Peng et al., 2024; Benara et al., 2024). TRUBRIC applies this to agentic traces by embedding a task-specific structured summary of each trace, preserving behavior-relevant information while minimizing the presence of system metadata.

Efficient benchmarking. Task-efficient benchmarking estimates a system’s full benchmark score from a small number of executions. The dominant paradigm for efficient benchmarking centers around item response theory (IRT) (Birnbaum, 1968; Embretson & Reise, 2000), where a target system’s score is estimated with $m \leq M$ item-level scores by assigning a latent ability estimated from the relationship between a task’s difficulty and discriminability based on $\mathcal{D}$ (Maia Polo et al., 2024; Kipnis et al., 2025). Extensions of IRT select informative or adaptive task subsets (Vivek et al., 2024; Truong et al., 2025) or apply it to agentic benchmarks directly (Ge et al., 2026; Ndzomga, 2026). A complementary paradigm represents the systems themselves, either from their score profiles (Ruan et al., 2024; Zhuang et al., 2025) or from the artifacts they produce while executing a task (Duderstadt et al., 2023; Acharyya et al., 2024; Helm et al., 2025). For example, the Product Kernel Perspective Space (PKPS) (Anonymous, 2026) represents language models as Euclidean objects obtained via multidimensional scaling of the distances between their embedded responses. Helm et al. (2026a) regress benchmark scores on similar representations and prove that the resulting predictions are task-efficient relative to the sample score. TRUBRIC enables extending geometry-based methods to agentic traces by providing an effective trace encoding. We show that these two approaches – score-based IRT and artifact-based PKPS – can be combined to outperform either for any number of given executed tasks.

5 DISCUSSION

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